

COURSE 2A

C2-00 SENIOR ANALYST FOUNDATIONS

BEGINNER-FIRST TEACHING BOOK

Optional Senior / BI Analyst path

Option B visual family

THIS BOOK IS YOUR TEACHER

This book is your teacher

Use one lesson at a time. New advanced concepts are still taught from first principles. Save a genuine attempt before opening protected review; do not skim ahead just because a topic name looks familiar.

Daily lesson loop

1) Why it matters -> 2) what you learn -> 3) prerequisite -> 4) picture it in your head -> 5) new words -> 6) watch / visual source when useful -> 7) follow with me once -> 8) expected result -> 9) why it worked -> 10) common beginner mistakes -> 11) recovery ladder -> 12) small fresh task -> 13) validate -> 14) teach it back -> 15) protected review + changed retry -> 16) evidence / mastery + exact next route.

Source rule

External videos are companions, not the course authority. If a useful source exists, open the exact link and return to the course task. If an external source would leak a solved capstone or add little value, the lesson explicitly uses an INTERNAL TEACHING ROUTE.

Attempt protection

Do the teacher example once. Then close it and complete the changed task from your own blank workbook/query/report/notebook. Protected review opens only after a saved attempt and is followed by a changed retry.

Mastery standard

You are finished only when you can reproduce the result, validate it, explain why it works, recover from a mistake and save the required evidence.

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Prerequisite and tool map

What Course 1 knowledge is assumed, what is new, and what files you use

ASSUMED FROM COURSE 1

Basic spreadsheet/database literacy; simple cleaning and filtering; basic SQL/Excel/Power BI outputs; ability to calculate and explain a simple metric; basic join awareness; saving evidence files. If any of these are not independent, repair Course 1 rather than turning C2-00 into a hidden replacement.

NEW IN C2-00

Decision ownership; scoping ambiguous requests; clarification meetings; metric contracts; explicit grain/source-of-truth/lineage reasoning; calibrated assumptions and limitations; professional validation/reconciliation; inheritable documentation. These are workplace-control skills, not harder button-clicking.

FILES YOU USE

1. This Teaching Book. 2. C2-00 Practice Workbook. 3. Mini-Lab brief + workspace. 4. Protected Practice Review only after genuine lesson attempts. 5. Gate A only after Mini-Lab. 6. B/C only when explicitly routed.

TOOLS

Excel or compatible spreadsheet tool for the workbook. No Power BI, Python, SQL server, Fabric tenant, or paid service is required for C2-00 mastery. Videos are support only. The concepts transfer to later BI and engineering work.

Practice, review and Gate route

The same attempt-first logic used across the five-course system



LESSON RULE

Teacher demo -> guided build -> independent fresh task -> validation -> explain-back. The protected lesson review stays closed until a genuine attempt is saved. After review, close it and solve a changed transfer task. The Mentor will later enforce the same rule in the common five-course state model.

MODULE RULE

After SAF1-SAF8 are mastered, complete the integrated Mini-Lab. Gate A is then the first independent competency Gate. Passing is competency-based: correct scope/definition/grain/validation/judgment/handoff with no unresolved critical failure.

RETEST RULE

A failed Gate does not unlock a solved answer. Repair only the weak competency, then use a changed task. Gate B and Gate C remain protected fresh full reassessments and appear only when the routing logic explicitly requires them.

SAF1 - Junior analyst vs senior analyst responsibility

1. Why it matters

A report can be technically correct and still damage a decision. Senior-level analysis means noticing what has not been defined, what could make the number wrong, and what another person needs in order to trust and reuse the work.

2. What you are learning

Separate task execution from decision ownership. Build a small ownership checklist that covers the decision, metric definition, grain/source, validation, risk communication, and handoff.

3. Prerequisite / what you already know

Course 1 basics: you can clean a small dataset, calculate a metric, and explain a chart. You do not need senior job experience.

4. Picture it in your head

Mental model

Imagine this as a visible path: Request -> Decision -> Definition -> Trust -> Handoff. Keep the stages separate. If the result becomes wrong, walk backward one stage at a time until you find where the meaning, rule, grain, relationship, or control changed.

5. New words before we use them

Decision owner: The person accountable for choosing the business rule or action. Acceptance criterion: A concrete check that tells you whether the analysis is good enough to use. Traceability: The ability to connect a conclusion back to definitions, source data, calculations, and decisions.

6. Watch / visual source

INTERNAL TEACHING ROUTE. No external video is required for this lesson. Use the mental model and worked demonstration below as the canonical teaching route.

7. Follow with me once

Teacher demonstration - do this once with me, then close the example before your fresh task.

- 1 Stakeholder says: 'Build the weekly sales dashboard by Friday.' A task-only response begins choosing charts.
- 2 First ask what decision Friday's meeting must make. Assume the answer is: decide whether to change the weekend promotion.
- 3 Write the minimum ownership contract: decision = promotion decision; metric = net revenue and margin; period = last 8 comparable weeks; grain = order line; exclusions = cancelled orders; sources = orders + returns + finance control.
- 4 Add one acceptance criterion: net revenue must reconcile to the finance weekly control within the agreed tolerance before recommendation.

8. What you should see / be able to say

Expected result

A short decision-protection checklist. It names the decision, definition, grain/source, validation check, and unresolved item before chart design begins.

9. Why it worked

Reasoning

The work is now judged by whether it supports a decision safely, not by whether a dashboard exists. Professional upgrade: In professional teams, senior analysts often spend more time reducing ambiguity, defining controls, documenting assumptions, and negotiating acceptance criteria than junior analysts expect.

10. Common beginner mistakes

1. Starting with chart colours or formulas. | 2. Treating the stakeholder's wording as a complete specification. | 3. Calling yourself 'senior' because you used an advanced tool. | 4. Hiding an unresolved definition so the work looks finished.

11. If you get stuck - recovery ladder

Recovery

Write one sentence: 'What decision could be wrong if my analysis is wrong?' Then list the definition, source, and validation most likely to affect that decision. If still stuck: (1) restate the decision; (2) fill only the required control fields; (3) use the partial template in the workbook; (4) save your genuine attempt before opening review.

12. Now you do a small version alone

Fresh independent task

Fresh scenario: Operations asks, 'Why are refunds high? Send me something today.' Create a six-line ownership checklist before doing any calculation.

13. Validate your result

Validation

A different analyst should be able to answer: What decision is being supported? Which metric definition is still open? What evidence would block delivery?

14. Teach it back in your own words

Explain without reading

Explain in 60-90 seconds why senior responsibility is not the same thing as using more advanced software.

15. Protected review + changed fresh retry

Attempt protection

Save your genuine attempt. Only then open the protected SAF1 review. Close it and repeat on a changed scenario without copying wording.

16. Evidence / mastery + exact next route

Mastery evidence

Ownership checklist + one validation criterion + 60-90 second explain-back. Master only when you can reproduce the pattern on a new request. After saving evidence, continue to SAF2.

SAF2 - Turning vague stakeholder requests into scoped analysis

1. Why it matters

Most real requests arrive incomplete: 'sales are down', 'customers are leaving', 'make a dashboard'. If you calculate before scoping, you can answer the wrong question perfectly.

2. What you are learning

Convert a vague request into an analysis brief: decision, audience, deadline, period, metric, grain, segments, exclusions, sources, and acceptance test.

3. Prerequisite / what you already know

SAF1 ownership mindset. Course 1 metric and filtering basics.

4. Picture it in your head

Mental model

Imagine this as a visible path: Vague request -> Decision -> Metric -> Grain -> Source -> Acceptance. Keep the stages separate. If the result becomes wrong, walk backward one stage at a time until you find where the meaning, rule, grain, relationship, or control changed.

5. New words before we use them

Scope: The agreed boundary of what the analysis will and will not answer. Comparison period: The time period used as the reference for saying something increased or decreased. Exclusion: A record or population deliberately left out for a stated reason.

6. Watch / visual source

Scoped analysis brief + unresolved-question list + acceptance test + explain-back. RECOMMENDED VIDEO - requirements interview 00:00-05:42: problem definition, gathering, refinement. Use for questioning technique; the book controls the course

<https://www.youtube.com/watch?v=iAbSOvf96E>

After watching/opening: return to this lesson and reproduce the worked example from your own file/query/report. Save course-task evidence, not a screenshot of the source.

7. Follow with me once

Teacher demonstration - do this once with me, then close the example before your fresh task.

- 1 Do not calculate yet. Ask: 'Which decision will this analysis support?' Assume: decide whether to reduce the September promotion.
- 2 Define the audience and deadline: Sales Director, Monday 10:00.
- 3 Define the metric: net revenue after returns, not gross order value. Comparison: September vs August and vs same month last year.
- 4 Declare grain and segments: order-line grain, region and product category. Exclude cancelled orders; keep returns as negative adjustments.
- 5 Choose sources and acceptance: order lines + return events; reconcile monthly net revenue to Finance before recommendation.

8. What you should see / be able to say

Expected result

A one-page analysis brief where a colleague could tell exactly what is being answered and what remains out of scope.

9. Why it worked

Reasoning

Each ambiguous word was converted into a decision or measurable contract before tools were used. Professional upgrade: A strong scope note also records what is explicitly out of scope and what would trigger a scope change. This protects both quality and delivery time.

10. Common beginner mistakes

1. Using 'sales' without defining gross vs net. | 2. Comparing unequal time windows. | 3. Adding every possible segment instead of the ones relevant to the decision. | 4. Letting the deadline force silent assumptions.

11. If you get stuck - recovery ladder

Recovery

Circle every vague noun or adjective in the request: sales, active, high, recent, best, churned. Replace each with a definition or an open question. If still stuck: (1) restate the decision; (2) fill only the required control fields; (3) use the partial template in the workbook; (4) save your genuine attempt before opening review.

12. Now you do a small version alone

Fresh independent task

Fresh scenario: Product says, 'Customers are churning after onboarding. Find the problem.' Build the scoped brief; do not calculate churn.

13. Validate your result

Validation

Someone who did not attend the request meeting must be able to restate the decision, population, time window, metric, and exclusions from your brief.

14. Teach it back in your own words

Explain without reading

Explain why 'clarify before calculate' saves time even when the stakeholder says the request is urgent.

15. Protected review + changed fresh retry

Attempt protection

Save the independent brief first. Review only after attempt, then rebuild using a changed metric or time window.

16. Evidence / mastery + exact next route

Mastery evidence

Scoped analysis brief + unresolved-question list + acceptance test + explain-back. RECOMMENDED VIDEO - requirements interview 00:00-05:42: problem definition, gathering, refinement. Use for questioning technique; the book controls the course <https://www.youtube.com/watch?v=iIAbSOvf96E>
After saving evidence, continue to SAF3.

SAF3 - Requirements gathering and clarification meetings

1. Why it matters

A meeting is useful only if it changes uncertainty into recorded decisions. Beginners often ask many questions but fail to record who approved what, by when, and what happens if no answer arrives.

2. What you are learning

Plan a short clarification meeting, ask only analysis-changing questions, capture owners and decisions, and leave unresolved items visible.

3. Prerequisite / what you already know

SAF2 scoped analysis brief.

4. Picture it in your head

Mental model

Imagine this as a visible path: Question -> Owner -> Answer -> Record -> Resolve / escalat.... Keep the stages separate. If the result becomes wrong, walk backward one stage at a time until you find where the meaning, rule, grain, relationship, or control changed.

5. New words before we use them

Elicitation: Drawing out information or requirements from stakeholders. Decision log: A dated record of questions, answers, owners, and unresolved items. Escalation: Routing an unresolved decision to the person with authority to decide it.

6. Watch / visual source

Agenda + decision log + owner map + read-back summary + explain-back. RECOMMENDED VIDEO - requirements interview 00:50-05:42: problem definition through refinement.

<https://www.youtube.com/watch?v=iAbSOvf96E>

After watching/opening: return to this lesson and reproduce the worked example from your own file/query/report. Save course-task evidence, not a screenshot of the source.

7. Follow with me once

Teacher demonstration - do this once with me, then close the example before your fresh task.

- 1 Prepare three questions only: Which revenue definition is approved? Are returns recognized on return date or original order date? Which regions are in scope?
- 2 For every question, write why the answer can change the analysis. This prevents low-value questioning.
- 3 During the meeting, record answer + owner + date. If the owner cannot decide, record the escalation owner.
- 4 End by reading back the decisions and open items. Ask: 'If these are correct, may I proceed with this scope?'
- 5 Send the decision log with the analysis brief. Do not rely on memory or chat history as the only record.

8. What you should see / be able to say

Expected result

A meeting agenda and decision log with questions tied to analytical impact, named owners, status, and deadline.

9. Why it worked

Reasoning

The meeting produced an auditable contract instead of a collection of informal comments. Professional upgrade: In high-pressure work, the analyst may propose a temporary assumption, but it must be labelled, approved by the right owner, and paired with a trigger for recalculation.

10. Common beginner mistakes

1. Asking questions whose answers cannot change the work. | 2. Recording an answer without the owner who approved it. | 3. Leaving 'TBD' items with no deadline. | 4. Treating silence as approval.

11. If you get stuck - recovery ladder

Recovery

For each open item write: decision needed, impact if unresolved, owner, deadline, temporary assumption allowed? yes/no. If still stuck: (1) restate the decision; (2) fill only the required control fields; (3) use the partial template in the workbook; (4) save your genuine attempt before opening review.

12. Now you do a small version alone

Fresh independent task

Fresh scenario: Marketing and Finance disagree on whether refunded orders count in campaign revenue. Prepare the agenda and decision log.

13. Validate your result

Validation

Every open question has an owner and a next action. Every resolved question is reflected in the scope or metric contract.

14. Teach it back in your own words

Explain without reading

Explain the difference between 'asking more questions' and 'reducing decision uncertainty'.

15. Protected review + changed fresh retry

Attempt protection

Attempt first. Then inspect the protected review and retry with a different stakeholder conflict.

16. Evidence / mastery + exact next route

Mastery evidence

Agenda + decision log + owner map + read-back summary + explain-back. RECOMMENDED VIDEO - requirements interview 00:50-05:42: problem definition through refinement. <https://www.youtube.com/watch?v=iIAbSOvf96E> After saving evidence, continue to SAF4.

SAF4 - KPI ownership, metric contracts and definitions

1. Why it matters

Two teams can show different numbers while both calculations are arithmetically correct. The disagreement is often a definition problem: different populations, dates, exclusions, or sources.

2. What you are learning

Build a versioned metric contract for a KPI and distinguish label, business meaning, formula, eligible population, exclusions, grain, time basis, owner, source, target, and tolerance.

3. Prerequisite / what you already know

SAF2 scope and SAF3 decision ownership.

4. Picture it in your head

Mental model

Imagine this as a visible path: Meaning -> Formula -> Population -> Time -> Owner -> Version. Keep the stages separate. If the result becomes wrong, walk backward one stage at a time until you find where the meaning, rule, grain, relationship, or control changed.

5. New words before we use them

KPI: A metric tied to an important business objective or decision. Denominator: The population used underneath a rate or percentage. Metric contract: A documented agreement defining exactly how a metric is interpreted and calculated.

6. Watch / visual source

Versioned metric contract + one reconciliation rule + explain-back. RECOMMENDED REFERENCE - IIBA KPI guidance Use for KPI framing; course contract fields are authoritative for assessment.

<https://www.iiba.org/professional-development/knowledge-centre/articles/kpis-for-business-analysts/>

After watching/opening: return to this lesson and reproduce the worked example from your own file/query/report. Save course-task evidence, not a screenshot of the source.

7. Follow with me once

Teacher demonstration - do this once with me, then close the example before your fresh task.

- 1 Net Revenue meaning: revenue recognized from completed order lines after approved returns and discounts.
- 2 Formula: $\text{sum}(\text{line_amount} - \text{approved_return_amount} - \text{discount_amount})$. Grain: order line. Time basis: order completion month.
- 3 Return Rate meaning: returned completed order lines divided by eligible completed order lines. State whether partial returns count by line or order.
- 4 Name exclusions: cancelled orders, test orders, fraudulent orders if approved. Name owner: Finance for revenue recognition; Operations may own operational return reason taxonomy.
- 5 Add version date and tolerance: e.g. reconcile monthly Net Revenue to Finance control within 0.5% or investigate before publishing.

8. What you should see / be able to say

Expected result

A metric contract where another analyst could implement the same KPI without guessing hidden rules.

9. Why it worked

Reasoning

The metric is treated as a business agreement first and a formula second. Professional upgrade: Organizations often separate metric owner, data owner, and report owner. Do not assume the dashboard developer has authority to define the business metric.

10. Common beginner mistakes

1. Writing only the formula. | 2. Ignoring the denominator of a rate. | 3. Using dashboard label as definition. | 4. Changing a metric silently when the source changes.

11. If you get stuck - recovery ladder

Recovery

If two numbers disagree, compare contract fields one by one: meaning, population, exclusions, date basis, grain, source, version. If still stuck: (1) restate the decision; (2) fill only the required control fields; (3) use the partial template in the workbook; (4) save your genuine attempt before opening review.

12. Now you do a small version alone

Fresh independent task

Fresh scenario: Define 'On-time Delivery Rate' for an executive dashboard. Include numerator, denominator, exclusions, time basis, grain, owner, source, target, and version.

13. Validate your result

Validation

Ask a second analyst to independently restate the numerator and denominator. If they cannot, the contract is not yet clear.

14. Teach it back in your own words

Explain without reading

Explain why a KPI can be wrong even when the formula contains no error.

15. Protected review + changed fresh retry

Attempt protection

Attempt first, then compare against protected signals and retry with a different rate metric.

16. Evidence / mastery + exact next route

Mastery evidence

Versioned metric contract + one reconciliation rule + explain-back. RECOMMENDED REFERENCE - IIBA KPI guidance Use for KPI framing; course contract fields are authoritative for assessment. <https://www.iiba.org/professional-development/knowledge-centre/articles/kpis-for-business-analysts/>
After saving evidence, continue to SAF5.

SAF5 - Grain, lineage and source-of-truth decisions

1. Why it matters

Many believable BI errors come from combining tables that describe different things: one row per order, one row per order line, one row per return event, one row per monthly target. If the grain is not declared, joins can multiply values silently.

2. What you are learning

Declare grain and keys, map lineage from source to output, and choose the source-of-truth owner for each business fact.

3. Prerequisite / what you already know

Course 1 joins/model basics plus SAF4 metric contract.

4. Picture it in your head

Mental model

Imagine this as a visible path: Source row -> Grain -> Transform -> Metric -> Report. Keep the stages separate. If the result becomes wrong, walk backward one stage at a time until you find where the meaning, rule, grain, relationship, or control changed.

5. New words before we use them

Grain: What exactly one row represents. Lineage: The trace of where data came from and how it changed before reaching the output. Source of truth: The agreed authoritative system or owner for a particular business fact.

6. Watch / visual source

Source/grain register + lineage map + join-risk note + explain-back. RECOMMENDED CURRENT REFERENCE - Power BI data lineage Use the lineage-view idea as a visual analogy; no Power BI service access is required for this lesson.

<https://learn.microsoft.com/en-us/power-bi/collaborate-share/service-data-lineage>

After watching/opening: return to this lesson and reproduce the worked example from your own file/query/report. Save course-task evidence, not a screenshot of the source.

7. Follow with me once

Teacher demonstration - do this once with me, then close the example before your fresh task.

- 1 orders: one row per order. Key = order_id.
- 2 order_lines: one row per order + line number. Key = order_id + line_id. Revenue is naturally calculated here.
- 3 returns: one row per return event. One line can have multiple return events, so joining directly without aggregation can duplicate line revenue.
- 4 targets: one row per region-month. It is a higher-grain target and should not be joined blindly to each order line for aggregation.
- 5 Write lineage: source tables -> cleaning rules -> return aggregation -> metric calculation -> executive report. State which system owns each fact.

8. What you should see / be able to say

Expected result

A source/grain register plus a small lineage chain. Every table has a declared row meaning and key.

9. Why it worked

Reasoning

You prevented fan-out by deciding what each row means before joining or aggregating. Professional upgrade: Source-of-truth is fact-specific. Finance can own recognized revenue while CRM owns customer segment and Operations owns fulfillment status.

10. Common beginner mistakes

1. Calling a file 'customer level' when it actually has multiple rows per customer. | 2. Choosing source of truth because it is easiest to access. | 3. Assuming a primary key without checking duplicates. | 4. Using DISTINCT to hide duplicated joins.

11. If you get stuck - recovery ladder

Recovery

Before any join, write two sentences: 'One row in left table means...' and 'One row in right table means...'. Predict the output grain before running the join. If still stuck: (1) restate the decision; (2) fill only the required control fields; (3) use the partial template in the workbook; (4) save your genuine attempt before opening review.

12. Now you do a small version alone

Fresh independent task

Fresh scenario: You have ticket, ticket_event, agent, and monthly_team_target tables. Declare grain/key and draw the lineage for 'resolved tickets per team vs target'.

13. Validate your result

Validation

For every join, state expected row count direction and one duplicate/unmatched-key check.

14. Teach it back in your own words

Explain without reading

Explain grain without using the word 'granularity'.

15. Protected review + changed fresh retry

Attempt protection

Save your source/grain register before opening the protected review. Fresh retry must use different table grains.

16. Evidence / mastery + exact next route

Mastery evidence

Source/grain register + lineage map + join-risk note + explain-back. RECOMMENDED CURRENT REFERENCE - Power BI data lineage Use the lineage-view idea as a visual analogy; no Power BI service access is required for this lesson.
<https://learn.microsoft.com/en-us/power-bi/collaborate-share/service-data-lineage> After saving evidence, continue to SAF6.

SAF6 - Assumptions, risks, limitations and analytical judgment

1. Why it matters

Decision-makers need to know not only what you found but how confident they should be. Mixing facts, assumptions, and recommendations makes a conclusion sound stronger than the evidence.

2. What you are learning

Label facts, assumptions, risks, limitations, and recommendations separately; calibrate recommendation strength to evidence.

3. Prerequisite / what you already know

SAF1-SAF5 decision, scope, metric, and source controls.

4. Picture it in your head

Mental model

Imagine this as a visible path: Fact -> Assumption -> Risk -> Limitation -> Recommendation. Keep the stages separate. If the result becomes wrong, walk backward one stage at a time until you find where the meaning, rule, grain, relationship, or control changed.

5. New words before we use them

Assumption: Something treated as true for the analysis but not fully proven. Risk: A possible future or hidden condition that could make the result wrong or harmful. Limitation: A known boundary of the data or method. Trigger: New evidence or a threshold that would change the recommendation.

6. Watch / visual source

INTERNAL TEACHING ROUTE. No external video is required for this lesson. Use the mental model and worked demonstration below as the canonical teaching route.

7. Follow with me once

Teacher demonstration - do this once with me, then close the example before your fresh task.

- 1 Fact: East net revenue is 12% below the approved target for the current month after reconciliation.
- 2 Assumption: the promotion calendar supplied by Marketing is complete.
- 3 Limitation: offline store orders arrive one day later and are not yet included.
- 4 Risk: if late offline orders are unusually high, the 12% gap may shrink materially.
- 5 Recommendation: investigate the promotion mix before expanding discounts. Trigger: recompute after offline close; reverse the recommendation if the gap falls below the agreed materiality threshold.

8. What you should see / be able to say

Expected result

A decision note where evidence strength is obvious and the reader knows what could change the conclusion.

9. Why it worked

Reasoning

The recommendation is neither timid nor overconfident; it is tied to evidence and a change trigger. Professional upgrade: Senior judgment often means choosing the safest decision language: 'evidence supports', 'directional only', 'do not act until', or 'act now with monitoring' based on consequence and uncertainty.

10. Common beginner mistakes

1. Calling an assumption a fact. | 2. Using 'probably' without naming the uncertainty. | 3. Listing limitations but ignoring them in the recommendation. | 4. Giving an irreversible action from weak evidence.

11. If you get stuck - recovery ladder

Recovery

Highlight every sentence in your recommendation. Label it F/A/R/L/REC. Any sentence you cannot classify needs rewriting. If still stuck: (1) restate the decision; (2) fill only the required control fields; (3) use the partial template in the workbook; (4) save your genuine attempt before opening review.

12. Now you do a small version alone

Fresh independent task

Fresh scenario: Customer satisfaction fell after a support-tool migration, but survey response rate also changed. Write fact/assumption/risk/limitation/recommendation + trigger.

13. Validate your result

Validation

A reviewer should be able to identify exactly which new evidence could strengthen, weaken, or reverse the recommendation.

14. Teach it back in your own words

Explain without reading

Explain why a limitation does not automatically make an analysis useless.

15. Protected review + changed fresh retry

Attempt protection

Attempt first. Retry on a scenario where the data limitation is different.

16. Evidence / mastery + exact next route

Mastery evidence

Calibrated decision note + trigger + explain-back. After saving evidence, continue to SAF7.

SAF7 - Professional validation and reconciliation

1. Why it matters

A dashboard can look polished while containing a duplicated join, missing period, stale file, or wrong denominator. Validation is not a final glance; it is a planned second path that tries to prove the result wrong.

2. What you are learning

Design validation before delivery using row counts, coverage, duplicates, unmatched keys, control totals, independent recomputation, business invariants, tolerances, and recovery.

3. Prerequisite / what you already know

Course 1 data-cleaning checks plus SAF4 metric contract and SAF5 grain.

4. Picture it in your head

Mental model

Imagine this as a visible path: Input checks -> Control total -> Independent recom... -> Invariant -> Release. Keep the stages separate. If the result becomes wrong, walk backward one stage at a time until you find where the meaning, rule, grain, relationship, or control changed.

5. New words before we use them

Reconciliation: Comparing results against an independent control or alternate calculation. Tolerance: The maximum acceptable difference before investigation is required. Invariant: A rule expected to remain true if the data and logic are correct.

6. Watch / visual source

Validation log + independent control + failure-recovery note + explain-back. RECOMMENDED CURRENT REFERENCE - Power BI data profiling Use the profiling section as visual reinforcement for detecting unexpected/null values; course validation plan is b

<https://learn.microsoft.com/en-us/training/modules/clean-data-power-bi/>

After watching/opening: return to this lesson and reproduce the worked example from your own file/query/report. Save course-task evidence, not a screenshot of the source.

7. Follow with me once

Teacher demonstration - do this once with me, then close the example before your fresh task.

- 1 Input check: confirm expected files and date coverage are present before calculating.
- 2 Key check: detect duplicate order-line keys and unmatched return events.
- 3 Control total: compare completed gross revenue to the operational source control.
- 4 Independent recompute: calculate Net Revenue a second way using a separate grouped summary or pivot, not the same formula copied twice.
- 5 Business invariant: Net Revenue should not exceed eligible gross revenue. Record tolerance and recovery steps for every failed check.

8. What you should see / be able to say

Expected result

A validation plan with expected values or control methods, observed result, tolerance, status, and recovery path.

9. Why it worked

Reasoning

Different failure types have different checks; one green total cannot prove every part of the pipeline. Professional upgrade: For important outputs, store validation evidence with the release so future analysts can see what was checked, what tolerance was accepted, and what failed.

10. Common beginner mistakes

1. Validating with the same formula that produced the result. | 2. Ignoring a small mismatch without an agreed tolerance. | 3. Fixing the data silently instead of recording the cause. | 4. Treating 'no error message' as validation.

11. If you get stuck - recovery ladder

Recovery

Ask: What wrong result could still look plausible? Design one check specifically to catch that failure. If still stuck: (1) restate the decision; (2) fill only the required control fields; (3) use the partial template in the workbook; (4) save your genuine attempt before opening review.

12. Now you do a small version alone

Fresh independent task

Fresh scenario: Validate a support dashboard showing resolved ticket rate by team. Design at least six checks before seeing the final dashboard.

13. Validate your result

Validation

At least one check must be independent of the main calculation and at least one must test grain/key integrity.

14. Teach it back in your own words

Explain without reading

Explain the difference between checking that code runs and checking that a business result is trustworthy.

15. Protected review + changed fresh retry

Attempt protection

Attempt first. After review, design a changed validation plan for a different metric and source structure.

16. Evidence / mastery + exact next route

Mastery evidence

Validation log + independent control + failure-recovery note + explain-back. RECOMMENDED CURRENT REFERENCE - Power BI data profiling Use the profiling section as visual reinforcement for detecting unexpected/null values; course validation plan is b <https://learn.microsoft.com/en-us/training/modules/clean-data-power-bi/> After saving evidence, continue to SAF8.

SAF8 - Documentation another analyst can inherit

1. Why it matters

If only you can refresh or explain the work, the team owns a fragile dependency. Senior-quality documentation lets another analyst reproduce the output and know when not to trust it.

2. What you are learning

Create an inheritable handoff pack covering purpose, definitions, sources/access, grain/keys, steps, validation, limitations, refresh/run instructions, owners, escalation, and change history.

3. Prerequisite / what you already know

SAF1-SAF7. You need one completed small analysis artifact to document.

4. Picture it in your head

Mental model

Imagine this as a visible path: Purpose -> Definitions -> Sources -> Steps -> Checks -> Limits -> Owner. Keep the stages separate. If the result becomes wrong, walk backward one stage at a time until you find where the meaning, rule, grain, relationship, or control changed.

5. New words before we use them

Runbook: Step-by-step instructions for operating or refreshing a repeatable process. Change log: A dated history of material changes and why they were made. Escalation owner: The person to contact when a rule, source, or process cannot be resolved by the analyst.

6. Watch / visual source

INTERNAL TEACHING ROUTE. No external video is required for this lesson. Use the mental model and worked demonstration below as the canonical teaching route.

7. Follow with me once

Teacher demonstration - do this once with me, then close the example before your fresh task.

- 1 Purpose: weekly executive check on net revenue, margin, and return rate; decision owner = Commercial Director.
- 2 Definitions: link to metric-contract version and state effective date.
- 3 Sources: exact source names, refresh timing, access owner, expected file/table grain.
- 4 Run steps: numbered sequence from source refresh through transformation, validation, report refresh, and evidence save.
- 5 Checks and limitations: list required reconciliations, tolerances, known delayed source, and what to do if a control fails.
- 6 Ownership: report owner, metric owner, data owner, escalation path, and change history.

8. What you should see / be able to say

Expected result

A new analyst can reproduce the normal run and recognize a failure without asking you what each step means.

9. Why it worked

Reasoning

The handoff documents decisions and failure handling, not just button clicks. Professional upgrade: Good documentation is concise but executable. Keep authoritative definitions in one controlled location and reference them instead of copying conflicting versions into many files.

10. Common beginner mistakes

1. Writing 'refresh data' without naming the source or expected timing. | 2. Documenting only the happy path. | 3. Leaving metric definitions inside your head. | 4. Putting passwords or secrets into documentation.

11. If you get stuck - recovery ladder

Recovery

Give the handoff to an imaginary analyst who has a clean folder and no chat history. Mark every sentence where they would need to ask you a question. If still stuck: (1) restate the decision; (2) fill only the required control fields; (3) use the partial template in the workbook; (4) save your genuine attempt before opening review.

12. Now you do a small version alone

Fresh independent task

Fresh scenario: Document a monthly customer-retention report so a new analyst can run it next month. Include failure and escalation paths.

13. Validate your result

Validation

Use the cold-handoff test: another person should identify inputs, outputs, definitions, steps, validations, limitations, and owners from the document alone.

14. Teach it back in your own words

Explain without reading

Explain why documentation is part of analytical quality rather than administrative work.

15. Protected review + changed fresh retry

Attempt protection

Attempt first, then compare against the protected review and rebuild one section from a changed failure scenario.

16. Evidence / mastery + exact next route

Mastery evidence

Handoff pack + cold-handoff checklist + explain-back. After saving evidence, complete the module mastery checklist and follow the Gate A / completion route.

C2-00 mastery checklist + quick reference

Use this for revision after the first full study pass

MASTER THESE EIGHT QUESTIONS

SAF1: What decision am I protecting? | SAF2: What exactly is in/out of scope? | SAF3: Which unanswered questions can change the analysis, and who owns them? | SAF4: Is the metric contract complete and versioned? | SAF5: What does one row mean, where did the data come from, and who owns the fact? | SAF6: Which statements are facts vs assumptions/risks/limitations? | SAF7: What independent check could prove my result wrong? | SAF8: Could another analyst reproduce and recover the work?

FINAL MODULE CHECK

For every lesson: genuine attempt saved; independent fresh task complete; validation evidence present; 60-90 second explain-back saved; protected review opened only after attempt; changed retry completed without copying; no unresolved critical ambiguity.

DO NOT CONFUSE

Completion percentage is not mastery. A polished dashboard is not trusted analysis. A stakeholder instruction is not automatically the source of truth. A matching total does not prove every join. A limitation is not permission to hide uncertainty. Documentation is part of quality.

COURSE POSITIONING

C2-00 belongs to optional Course 2A. It deepens workplace judgment after Course 1. A learner following the direct DA -> DE route may skip Course 2A and use Course 2B after meeting Course 1 exit competencies. C2-00 is valuable enrichment, not a hidden prerequisite for C2B or Course 3.

C2-00 source / video registry

External media supports visual learning; the internal book is sufficient if every link fails

SAF2 / SAF3 - RECOMMENDED VIDEO

How I Conduct A Requirements Gathering Interview As A Business Analyst

00:00-05:42

<https://www.youtube.com/watch?v=ilAbSOvf96E>

Title and chapter metadata resolved on web 2026-09-10.

SAF4 - RECOMMENDED REFERENCE

IIBA - KPIs for Business Analysts

KPI framing; no course task depends on it

<https://www.iiba.org/professional-development/knowledge-centre/articles/kpis-for-business-analysts/>

Page reachable on web 2026-09-10.

SAF5 - RECOMMENDED CURRENT REFERENCE

Microsoft Learn - Power BI data lineage

Use lineage view as visual analogy

<https://learn.microsoft.com/en-us/power-bi/collaborate-share/service-data-lineage>

Current page found on web 2026-09-10; page notes its embedded video may show an earlier UI.

SAF7 - RECOMMENDED CURRENT REFERENCE

Microsoft Learn - Clean, transform, and load data in Power BI

Use profiling section as visual reinforcement

<https://learn.microsoft.com/en-us/training/modules/clean-data-power-bi/>

Current training module found on web 2026-09-10.

NO FILLER / NO HIDDEN DEPENDENCY

SAF1, SAF6 and SAF8 intentionally remain book-led. C2-00 does not need eight separate videos to look 'video-rich'. A source is assigned only when it materially improves visualization or professional context. Assessment answers are never outsourced to external media.